CHT PLATFORM · DATA GOVERNANCE

Data Dictionary & Metric Definitions

What each reported number means, where it comes from, how it is calculated, how often it refreshes and where it is unreliable. Covers Platform, Content Hub, Reports and Companion.

How to read this document

Section 2 is the one-page answer: every metric in the same six columns (Metric, Source, Definition, Calculation, Refresh, Known limitation). Section 3 shows how data moves between systems. Section 4 is the field-level dictionary for each repo. Section 6 proposes the definitions we should standardise on.

Version	1.0 (draft for product and data review)
Scope	`develop` branch of each repo, read-only static review
Platform	`cht-platform-tool` @ `ad32e73` (2026-10-09)
Content Hub	`cht-content-hub` @ `72d245d` (2026-10-09)
Reports	`cht-reports` @ `a641074` (2026-10-08, template 1.2.0)
Companion	`cht-companion` @ `b9a0e7e` (2026-09-30)
Method	Each definition is traced to the code that computes it (file:line in grey). Items marked *inferred* depend on third-party API behaviour or runtime ordering that code alone cannot prove.
Classification	Confidential. Internal use only.

Contents

1 Summary
2 Metric definitions
3 Data lineage
4 Data dictionary by repo
5 Refresh schedule
6 Conflicts and recommended definitions
7 Known limitations (consolidated)
8 Glossary

1 Summary

Most client reporting depends on three words: **views**, **engagement** and **attendance**. In the current code, "views" has five different meanings, "engagement" is not defined anywhere, and "attended" means "joined the Zoom session at least once, for any length of time". Every other number in the reports builds on these three, so they need to be defined first.

VIEWS ARE PLATFORM-NATIVE LIFETIME TOTALS

YouTube `viewCount` and clip `view_count` are cumulative counts as of the last sync, not views inside the report window. No system collects watch time. The YouTube field labelled "average view duration" actually holds the video's length.

ENGAGEMENT IS UNDEFINED

No repo computes an engagement count or rate. The only computed ratios are LinkedIn CTR and video completion rate. The funnel stage named "Engaged" is HubSpot landing-page views.

ATTENDANCE MEANS ANY JOIN

One Zoom join event of any duration counts as attended and verifies the registrant. If a Zoom report import exists for a program, it replaces all webhook data for that program. Attendees seen only through webhooks have no duration, so they count as 0 minutes watched.

REFRESH IS MOSTLY ON DEMAND

In prod, Content Hub has no scheduled refresh: the weekday 14:00 UTC export ingest runs in dev only. YouTube and LinkedIn sync when an admin clicks Sync. Meta, livestream and survey platform data arrive by CSV upload. Nothing refreshes "daily" today.

REPORTS COMPUTE FEW NUMBERS IN CODE

cht-reports computes attendance rate, attendee breakdowns and survey percentages itself. Platform performance (YouTube, LinkedIn, Meta) and HubSpot data are passed to the model as raw text, so the numbers in the narrative are not calculated deterministically.

SOME SCREENS DISAGREE

Examples: two different "total earnings" figures; video "complete" at 95% in the database but 100% in GA4; survey unlock after 60 minutes for some checks and 105 minutes for others; the funnel counts registration rows, not distinct people.

2 Metric definitions

Status pills: **Computed** calculated in our code; **Native** passed through from a third-party API as-is; **Not defined** no implementation exists; **Proposed** see Section 6. File references are relative to each repo's backend source root.

Metric	Source	Definition	Calculation	Refresh	Known limitation
VIEWS					
Views (YouTube, campaign) Native	YouTube Data API v3 `videos.list` `statistics.viewCount`, via Content Hub connector	Platform view count for each campaign video published in the reporting period (or each video ID listed explicitly in the campaign settings)	Native value per video, no CHT math hub `services/connectors/youtube.py:197–209`	On demand (admin Sync in Content Hub). One snapshot per campaign per UTC day.	Lifetime total, not views inside the window. YouTube's own counting rules apply. No watch time (needs the YouTube Analytics API, not implemented). The "average view duration" column holds video length.
Clip views (catalog) Computed	Content Hub `posts` (YouTube and podcast posts synced from ops-console and MediaHub)	Total views of an official CHM clip (`channel = chm-official`) across all its posts	`view_count = Σ posts.view_count` per clip, at request time. Likes and comments are summed the same way. hub `public/clips.py:332–334`; platform cache `contenthub-catalog.service.ts`	Whenever ops-console pushes post stats (schedule owned outside these repos). Platform caches for up to 1 hour.	Lifetime. Mixes platforms. Neither repo controls freshness. The `/clips` sort has no tiebreaker, so a clip can appear on two pages.
Video views (LinkedIn) Native	LinkedIn Marketing API `adAnalytics` (version 202501), pivot CAMPAIGN	LinkedIn-reported video views of sponsored campaign content in the period	Native `videoViews`, one row per day in the period hub `services/connectors/linkedin_ads.py:95–111`	On demand (admin Sync)	Paid (sponsored) only, no organic posts. LinkedIn's view threshold applies (inferred).
Views (Meta, livestream)	CSV export uploaded to Content Hub	Whatever the uploaded export reports	No CHT math. Rows stored as uploaded. hub `campaign_platform_data.rows` (source=upload)	Manual upload	No API connector. Column names and definitions depend on the export used.
On-platform video watch Computed	Platform browser player (`WatchVideo`)	Per-learner progress on CHT program videos	`watchedSeconds` +1 per second while the tab is visible. `progress = watched / duration` (0 to 1). Complete at `progress ≥ 0.95`. platform `fe/pages/WatchVideo.tsx:14, 80–123`	Client sync every 15 s and on tab hide	Visible-tab timer, not player time (a paused but visible video still counts). Values are reported by the client. API and schema document progress as 0–100, but the client sends 0–1.

Metric	Source	Definition	Calculation	Refresh	Known limitation
Clip page views (web)	GA4 via GTM `GTM-MR68PMPZ`, event `clip_view`	Public clip page loads on the CHT site	GA4 event count platform `fe/lib/analytics.ts:31-46`	Real time in GA4	Not stored in any CHT database and not used in reports.
ENGAGEMENT					
Engagement (overall) Not defined	None	No repo defines an engagement count or rate	None. The `engagement_rate` column in Hub's `clip_analytics` table has no code that writes to it.	None	Needs a documented formula. See the proposal in Section 6. Proposed
CTR (LinkedIn) Computed	LinkedIn Ads API	Clicks per impression on sponsored content	`clicks ÷ impressions`, 4 significant figures; 0 if impressions = 0 hub `linkedin_ads.py:110`; `common.py:52`	On demand	Paid only. Stored as text. A rate of 0 can mean "no impressions".
Video completion rate (LinkedIn) Computed	LinkedIn Ads API	Share of video views that played to the end	`videoCompletions ÷ videoViews` hub `linkedin_ads.py:108`	On demand	Daily rates cannot be averaged. Re-derive the rate from the summed numerator and denominator.
Dwell time, impressions, clicks (LinkedIn) Native	LinkedIn Ads API	Average seconds in view; times shown; clicks	Native `averageDwellTime`, `impressions`, `clicks`	On demand	LinkedIn API definitions apply. Paid only.
Likes, comments (clips) Computed	Content Hub `posts`	Total reactions on a clip across its posts	`Σ like_count`, `Σ comment_count` hub `public/clips.py:333-334`	As for clip views	`share_count` and `impression_count` are stored on posts but not exposed by any API or report.
Funnel "Engaged" stage Computed	HubSpot campaign metrics (Platform admin funnel)	Landing-page views for the selected campaigns	Sum of every numeric HubSpot field named `views` platform `campaigns-funnel.service.ts:197`; `hubspot-campaign-metrics.util.ts:40`	On demand. Default window: last 90 days.	The generic key search can double count. The unit (page views) differs from the CHT stages (registrations).
Email sent, opens, clicks (HubSpot) Native	HubSpot email statistics	Marketing email performance per campaign	Summed per campaign row, then across rows platform `campaigns-dashboard.service.ts:290-340`	On demand	The statistics endpoint is portal-wide (inferred) and is merged into every campaign row, then summed. Totals can be multiplied by the number of campaigns.
ATTENDANCE					

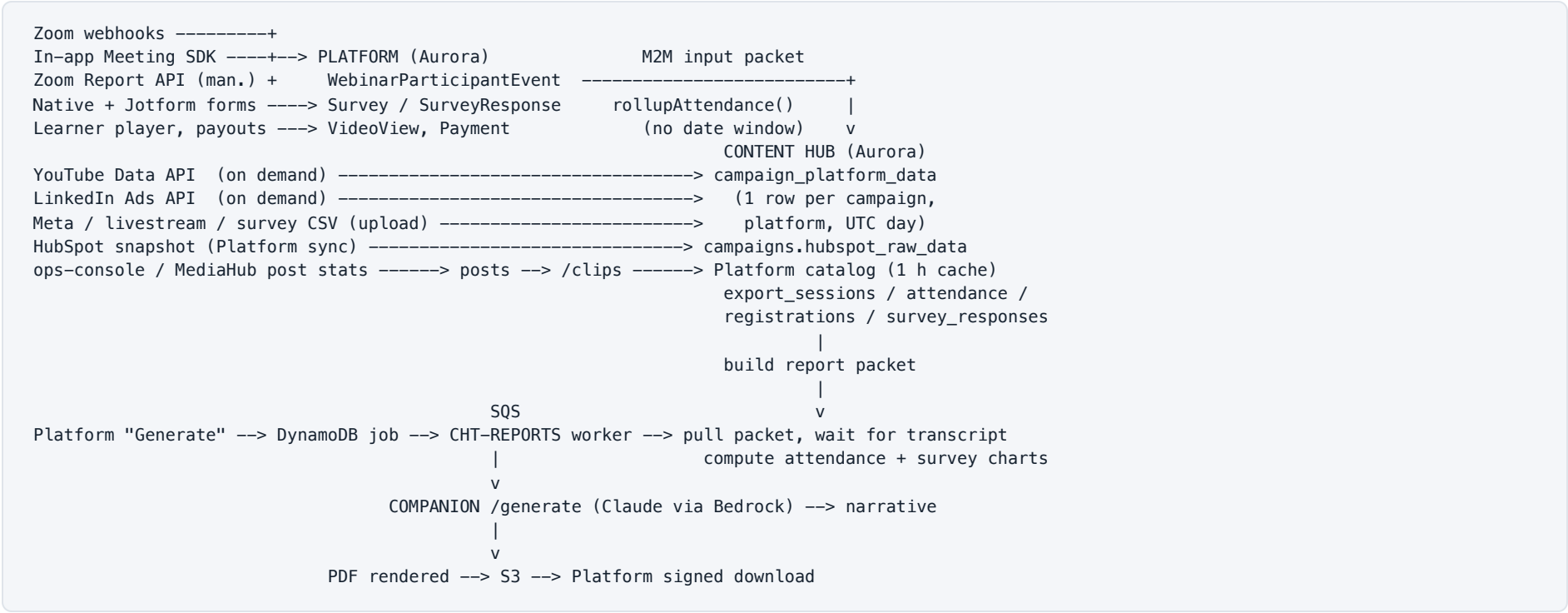
Metric	Source	Definition	Calculation	Refresh	Known limitation
Registered Computed	Platform `ProgramRegistration`, copied to Hub `export_registrations`	Registrations for the campaign's linked programs that are not rejected	`count(export_registrations)`. Includes PENDING, WAITLISTED and SURVEY_SUBMITTED. platform `export.service.ts:156`; hub `services/report_packet.py:674`	Hub ingest on link, transcript pull, report generate or refresh (weekday 14:00 UTC schedule in dev only)	Not limited to the report date window. Includes registrants who were never approved.
Attended Computed	Zoom webhooks, in-app Meeting SDK joins and the Zoom Report API, rolled up by Platform and sent to Hub	Distinct people (by email, otherwise user ID) with at least one Zoom join, excluding hosts and panelists	Platform combines each person's join rows into one. If any Report API rows exist for the program, only those are used. Hub counts the combined rows. platform `export.service.ts:490–566`; hub `report_packet.py:666–676`	Zoom events arrive in real time. Report import is manual. Hub picks them up on the ingest schedule.	No minimum minutes. A partial report import discards webhook attendees. Webhook-only hosts are removed only by the panelist email list. **Reconciliation required.**
Attendance rate Computed	cht-reports	Share of registrants who attended	`round(attended ÷ registered × 100)` reports `render/report-doc.service.ts:207`	Per report run	The denominator includes pending and waitlisted registrants. The rate can exceed 100% if people attend without registering.
Avg minutes watched Computed	Hub report packet, then cht-reports	Mean minutes in session per attendee	Hub: `Σ floor(duration_s / 60) ÷ attended`, 1 decimal place. Reports rounds to a whole number. hub `report_packet.py:654–681`; reports `content/attendees.ts:50`	Per report run	Attendees without a duration (webhook-only) count as 0 minutes, which understates the mean. Each attendee is floored, then the mean is rounded again.
Attendees by specialty, institution Computed	Platform `User.specialty` and `institution`, via the packet	Count of attendees per profile value	Group and count. Institutions: top 9 plus "Other institutions". Blank values are shown as "Not provided". reports `content/attendees.ts:11–55`	Per report run	Profile values are self-reported and free text, so the same institution can appear under different spellings.
Verified attendance (Platform) Computed	Platform `postEventAttendanceStatus`	Attendance accepted for survey and honorarium	VERIFIED if the registrant's email matches any join row after the session ends, or if an admin sets it platform `webinars/zoom-attendance-match.ts:105–143`	Zoom "ended" webhook. Report import (auto-verify is off by default). Admin action.	No minimum duration. An in-app SDK join verifies the user's own account. The webhook skips signature checks when the signature header is missing.
Funnel Registered, Attended, Converted Computed	Platform	Approved, then verified, then survey acknowledged, within the window	Count of registration rows by status and date platform `campaigns-funnel.service.ts:1379–1420`	On demand	Counts rows, not distinct people. Auto-approved rows are dated by `updatedAt`, which changes on later edits. NOT_REQUIRED attendance never counts as attended.

Metric	Source	Definition	Calculation	Refresh	Known limitation
No-show Not defined	None	None	None. Non-joiners stay PENDING_VERIFICATION.	None	No count or rate exists. Proposed
SURVEYS					
Survey responses Computed	Platform native forms and the Jotform webhook, copied to Hub `export_survey_responses`	One response per user per survey	Count of rows. FEEDBACK surveys keep the first submission; other types keep the latest. platform `surveys.service.ts:1600–1643`	On submit. Reaches Hub on the ingest schedule.	A Jotform submission without the hidden `user_id` field is dropped. Legacy Jotform forms shared by several programs attach every submission to each program (1,000-row cap).
Survey completion rate Computed	Platform admin analytics	Respondents per approved registrant	`round(uniqueRespondents ÷ APPROVED registrations × 100)` platform `utils/survey-analytics.ts:458–466`	On demand	The denominator includes people who could not open the survey (denied or unverified). The rate can exceed 100%.
Answer distribution (report) Computed	cht-reports	Share of respondents choosing each option	`round(count ÷ responses × 100)` . FEEDBACK surveys only, up to 12 options, contact fields removed. reports `render/report-doc.service.ts:256`	Per report run	For multi-select questions, percentages add up to more than 100%.
Survey score	Learner client	Numeric score	Stored exactly as the client sends it. Null for Jotform.	On submit	No server-side scoring or answer key.
HONORARIA AND EARNINGS					
Honoraria paid Computed	Platform `Payment` (Stripe or Bill.com)	Payouts that reached PAID	`count` and `Σ amount` where `status = PAID` , in cents platform `admin.controller.ts:286–295`	On demand. Status changes come from payout-provider webhooks.	The worker only creates PENDING rows. An admin must click "Pay now".
Total earnings (HCP)	Platform	Lifetime amount paid to an HCP	Dashboard: `Σ PAID` payments. Settings and summary: the `User.totalEarnings` counter. platform `dashboard.service.ts:43–45, 302`	On demand / on PAID	The two figures can drift apart. The counter is not adjusted on admin deletes or Bill.com failures, and webhook ordering can double count it.
Pending payments	Platform	Not yet paid	`Σ amount` where status is PENDING or PROCESSING	On demand	The Earnings page counts only the last 100 rows. The field named `availableBalance` is actually Σ PAID.
LEARNING AND CME					

Metric	Source	Definition	Calculation	Refresh	Known limitation
Program completion	Platform `ProgramEnrollment`	Every video in the program is complete	The learner has a view on every video and each view has `completed = true` (each at least 95% watched) platform `programs.service.ts:574–659`	On video-progress sync	Live webinars never mark completion. GA4 counts "complete" at 100%; the database uses 95%.
CME credits earned	Platform	Credits from completed programs	`Σ Program.creditAmount` over completed enrollments platform `dashboard.service.ts:117–123`	On demand	The `CMECredit` table is never written (the CME worker is disabled). Webinars earn 0 credits.
CAMPAIGN MARKETING AND REPORT INPUTS					
HubSpot campaign totals Native	HubSpot Marketing API (Platform dashboard). Content Hub stores an unparsed snapshot.	Sessions, influenced contacts, new contacts, form submissions, social clicks, event registrations	Recursive search through the HubSpot JSON, summing matching numeric keys platform `hubspot-campaign-metrics.util.ts:30–113`	On demand. Platform can also save a cached snapshot.	Matching keys by name can double count. Content Hub passes the snapshot to the model without calculating anything.
Input completeness	Hub report packet	Whether each of the 10 report inputs is present	Present or missing for HubSpot, LinkedIn, Meta, YouTube, sessions, attendance, registrations, KOLs, survey responses and template hub `services/report_packet.py`	Per packet	Checks presence only, not freshness or date window.
Narrative figures	Claude Sonnet via Companion `/generate` (Bedrock)	Numbers quoted in the 13 narrative sections	Written by the model from raw platform and HubSpot text. Transcript truncated at 100k characters; output capped at 16k tokens. reports `reports/orchestrator.ts:258`	Per report run	Not computed by code, so numbers can vary between runs. Should be limited to figures already in the packet.
COMPANION (OPERATIONAL)					
Chat and generate telemetry	Companion structured logs (CloudWatch)	Citations, retrieval time, latency, token counts, finish reason	Log events: `chat_retrieval`, `chat_done`, `bedrock_generate_ok`	Per request	Logs only. Not stored in a database and not shown on any dashboard. No user-facing content metrics.

3 Data lineage

Data enters through four routes: Zoom and learner activity into Platform; Platform's packet into Content Hub; third-party platform data into Content Hub; and post statistics from ops-console into Content Hub. Reports reads only the Content Hub packet.



Content Hub refreshes the Platform export when it is triggered by a campaign link, the first transcript pull, report generate, Refresh data, or (dev only) the weekday 14:00 UTC schedule. Attendance verification and survey submissions do not trigger a refresh by themselves.

4 Data dictionary by repo

4.1 Platform (`cht-platform-tool`)

Table.field	Type	Meaning
`Program.zoomSessionType`	WEBINAR | MEETING	Live session kind (meeting = office hours). Null for on-demand video programs.
`Program.startDate` , `duration`	DateTime, Int (min)	Scheduled start and length. Used for survey unlock when Zoom has not sent an end time.
`Program.zoomSessionEndedAt`	DateTime?	Actual end time from the Zoom "ended" webhook. Only ever moves later.
`Program.campaignId`	String?	Linked Content Hub campaign (numeric ID stored as text). Defines what goes into the input packet.
`Program.honorariumAmount` , `creditAmount`	Int cents, Float	Honorarium offered; CME credits offered.
`ProgramRegistration.status`	enum	SURVEY_SUBMITTED, PENDING, APPROVED, REJECTED, WAITLISTED. One row per user per program.
`ProgramRegistration.postEventAttendanceStatus`	enum	NOT_REQUIRED, PENDING_VERIFICATION, VERIFIED, DENIED. Gates the survey and honorarium.
`ProgramRegistration.postEventSurveyAcknowledgedAt`	DateTime?	FEEDBACK survey completed. Defines "Converted" in the funnel.
`WebinarParticipantEvent`	row	One row of Zoom join or leave evidence: `event` (JOINED/LEFT), `source` (WEBHOOK/REPORT_IMPORT; SDK rows are stored as WEBHOOK), `participantEmail` , `userId` , `isHost` . `joinTime` , `leaveTime` and `durationSeconds` are filled only by report import.
`Survey.type`	enum	PRE_TEST, POST_TEST, FEEDBACK (post-event), INTAKE (registration). `jotformFormId` set means the survey is Jotform-backed.
`SurveyResponse`	row	`answers` JSON, `score` (client-supplied), `submissionId` , `submittedAt` . Unique per user per survey.
`VideoView`	row	`watchedSeconds` , `progress` (sent as 0–1, documented as 0–100), `completed` , `completedAt` .

Table.field	Type	Meaning
`ProgramEnrollment`	row	`overallProgress` (mean of video progress), `completed`, `completedAt`.
`Payment`	row	`amount` (cents), `type` (HONORARIUM, SURVEY_BONUS; CME_COMPLETION and REFERRAL unused), `status` (PENDING, PROCESSING, PAID, FAILED), `paidAt`, `idempotencyKey`.
`User.totalEarnings`	Int cents	Running counter of PAID amounts. Can drift from the Payment rows.
`User.specialty`, `institution`	String?	Self-reported profile values. Feed the report's attendee breakdowns.
`CMECredit`, `AnalyticsEvent`, `User.lastLoginAt`	row / field	Defined in the schema but never written.

PLATFORM INPUT PACKET (SENT TO CONTENT HUB)

`GET /api/export/reports/campaigns/:campaignId/input-packet`, Cognito M2M scope `platform/export.read`. Includes every program linked to the campaign, with no date filter.

Block	Fields and rules
`sessions[]`	`platformToolProgramId`, `kind`, `title`, `sessionDate` (program start, otherwise recording start), `zoomMeetingId`, `zoomMeetingUuid`, `transcriptS3Key`, `transcriptStatus` (ok/missing), `chmProgramId`.
`attendance[]`	One row per person: `participantEmail`, `userId`, earliest `joinTime`, latest `leaveTime`, summed `durationSeconds` (null if 0), `source`, `specialty`, `institution`, `platformEventId` = `rollup:{program}:{email or user}`. Uses JOINED rows only. Uses report rows only, if any exist. Hosts and panelists are dropped.
`registrations[]`	`userId`, `registeredAt`, `status` (all except REJECTED), `specialty`, `institution`.
`surveys[]`	`surveyId`, `type`, `source` (native/jotform), `questions[]`, `responseCount`, `responses[]`. Programs that only have a Jotform URL are fetched live as `legacy-{type}:{formId}`.

4.2 Content Hub (`cht-content-hub`)

Table	Key fields	Meaning / writer
`export_sessions`	`platform_tool_program_id`, `campaign_id`, `kind`, `session_date`, `zoom_meeting_id`, `transcript_s3_key`, `transcript_text`, `zoom_session_ended_at`	Platform sessions. Written by export ingest.

Table	Key fields	Meaning / writer
`export_attendance_events`	`dedupe_key`, `event` (defaults to JOINED), `source`, `participant_email`, `duration_seconds`, `join_time`, `leave_time`, `user_id`, `specialty`, `institution`	One row per rolled-up attendee. Written by export ingest.
`export_registrations`	`user_id`, `registered_at`, `status`, `specialty`, `institution`	Platform registrations (not rejected). Written by export ingest.
`export_survey_responses`	`respondent_id`, `source`, `survey_type`, `submitted_at`, `answers`, `questions`	Platform survey responses. Written by export ingest.
`export_ingest_runs`	trigger, status, counts	Audit of each Platform-to-Hub pull.
`campaign_platform_data`	`campaign_id`, `platform`, `fetch_date` (UTC), `status`, `rows` JSON, `source` (api/upload), `synced_at`, `next_sync_at`	One row per campaign, platform and UTC day; same-day syncs overwrite. Holds YouTube, LinkedIn, Meta, livestream and survey platform data. `next_sync_at` is set to the next day, but nothing schedules it.
`platform_sync_runs`	platform, status, error	Audit of each connector sync.
`posts`	`clip_id`, `platform`, `posted_at`, `view_count`, `like_count`, `comment_count`, `share_count`, `impression_count`, `stats_synced_at`, `source` (webhook/direct)	Platform posts with engagement counters. Written by ops-console and MediaHub; nothing in this repo writes them.
`clips`, `shoots`, `kols`, `campaign_kols`	catalog and KOL fields	Content catalog and KOL roster that feed the report's KOL section.
`reports.clip_analytics`, `reports.campaign_analytics`	incl. `engagement_rate`	Warehouse tables with no ingest code. Unused.

Report packet attendance block: `registered_count` = number of registration rows; `attended_count` = number of JOINED attendee rows; `avg_minutes_watched` = mean of floored minutes (0 when duration is missing); `attendees[]` = specialty, institution and minutes, with no names.

4.3 Reports (`cht-reports`)

Item	Definition
Report job	DynamoDB item owned by Platform: `campaign_id` , `report_id` , `template_type` (executive_summary 1.0.0 / 1.1.0 / 1.2.0), `window_start` , `window_end` , `status` , `s3_key_pdf` , `version` .
Lifecycle	queued, pulling_data, waiting_for_transcript (re-checked every 300 s for up to 180 min), generating, rendering, uploading, complete or failed. SQS allows up to 3 receives.
Computed in code	Registered and attended (passed through; attended falls back to the attendee count), attendance rate, average minutes (rounded), attendees by specialty and institution, survey answer percentages.
Passed to the model	Transcript (up to 100k characters), platform slices (YouTube, LinkedIn, Meta rows), HubSpot snapshot, KOLs (from Hub, or extracted from the transcript as a fallback), session format (inferred from Q&A markers).
Output	13 narrative sections plus code-built charts, rendered to a PDF on S3. Users only ever receive the PDF.

4.4 Companion (`cht-companion`)

Table / item	Key fields	Meaning
`sources`	`source_type` (youtube_caption, catalog_clip, curated_doc), `status` (pending, approved, rejected, soft_deleted), `doctors[]` , `topics[]` , `content_date` , `chunk_count` , `embedding_model`	Knowledge-base documents. Only `approved` sources are retrievable.
`chunks`	`text` , `embedding` vector(1024) (Titan v2), `speaker` , `timestamp_start` , `timestamp_end` , `url`	Retrieval units. HNSW cosine index (m 16, ef_construction 128).
`ingest_jobs`	`kind` (initial, reindex, refresh), `status` , `chunks_before` , `chunks_after` , `chunks_skipped`	Ingest audit.
Retrieval	k = 10, sibling radius 1	On develop there is no reranker and no distance cutoff.
Rate limits	chat 20/min, 500/day; generate 5/min, 200/day	Per caller.
Ingest coverage	youtube_caption discovery runs daily	Caption downloads fail with 403, so nothing is queued. `catalog_clip` exists in the schema only; `curated_doc` is added manually.

5 Refresh schedule

Job / feed	Repo	Schedule	Prod	Feeds
Zoom webhooks	Platform	Real time	On	Join and leave rows, session end, auto-verify
Zoom attendee report import	Platform	Manual (admin)	On	Durations and the authoritative attendee list
Session reminders	Platform	`cron(0 0/3 * * ? *)` by default; every 6 h on platform, every 12 h on dev	On (6 h)	24-hour reminder emails
Platform export ingest	Content Hub	`cron(0 14 ? * MON-FRI *)`, 25 campaigns per batch, plus event triggers	Off (triggers only)	`export_*` tables
YouTube / LinkedIn sync	Content Hub	On demand (admin Sync or Sync all)	On	`campaign_platform_data`
Meta, livestream, survey uploads	Content Hub	Manual CSV upload	On	`campaign_platform_data`
Post statistics	external (ops-console)	Not defined in these repos	External	`posts`, so clip views, likes and comments
HCP intel poll	Content Hub	`rate(30 minutes)`	Off (also off in dev)	KOL activity (PubMed, CT.gov, news, Bluesky, CMS, Open Payments, NIH)
OpenAlex	Content Hub	`cron(30 3 ? * SUN *)`	Off (also off in dev)	KOL publications
KOL–HCP matcher	Content Hub	`cron(0 4 * * ? *)`	Off (also off in dev)	KOL to HCP links
Post tagging / playlist doctor tagger	Content Hub	`rate(12 hours)` / `cron(30 4 * * ? *)`	Off (tagging on in dev)	Clip and playlist tags
WordPress projection ops	Content Hub	`cron(0 3 * * ? *)`; ingest via SQS	Ingest on; nightly reconcile Off	WordPress catalog
Report generation	Reports	On demand (Generate or Regenerate)	On	PDF

Job / feed	Repo	Schedule	Prod	Feeds
Companion caption discovery	Companion	Daily	On (blocked)	Knowledge base (currently blocked by caption 403s)
Platform catalog cache	Platform	Redis TTL 1 h (clips, YouTube), 5 min (WordPress)	On	Public catalog pages

6 Conflicts and recommended definitions

The table below lists each place where one word currently has several meanings, followed by a proposed canonical definition. Items marked Proposed need product sign-off before engineering changes the code.

Term	Meanings in code today	Proposed canonical definition Proposed
Views	(1) Hub clip `view_count`, summed over posts; (2) YouTube `viewCount`; (3) HubSpot landing-page "views"; (4) GA4 `clip_view`; (5) `VideoView` progress rows.	**Views** = native platform views for the content, reported *per platform* and never summed across platforms without a label. State "lifetime as of <sync date>" unless a window delta is computed. Call HubSpot's figure "page views" and the Platform figure "watch progress".
Engagement	Not defined. The funnel "Engaged" stage is HubSpot landing-page views. LinkedIn has CTR, completion rate and dwell time.	**Engagements** = likes + comments + shares + clicks (whichever the platform reports). **Engagement rate** = engagements ÷ impressions; if impressions are not available, engagements ÷ views, labelled as such. Rename the funnel stage to "Visited".
Attended	(1) Any JOINED row; (2) verified status; (3) report-import segment count; (4) "seen in Zoom" (email or user ID).	**Attended** = distinct registrant with at least *N* cumulative minutes in the live session (suggest N = 10, or 25% of scheduled length), using report-import data when available and webhook data otherwise, merged per person rather than all-or-nothing. Hosts and panelists excluded.
Registered	Report: all non-rejected rows, no window. Funnel: APPROVED within the window.	**Registered** = APPROVED registrations for the program. Report pending and waitlisted counts separately.
Attendance rate	attended ÷ all non-rejected registrations.	Attended registrants ÷ approved registrants. Report walk-ins (attended, not registered) separately so the rate cannot exceed 100%.
Avg minutes watched	Missing durations count as 0; floored, then rounded.	Mean over attendees *with a known duration*, rounded once, and the coverage shown ("n of N attendees had duration data").
No-show	Not defined.	Approved registrants who did not attend after the session ended. **No-show rate** = no-shows ÷ approved.
Video complete	95% in the database; 100% (YouTube ENDED) in GA4.	95% everywhere. Rename the GA4 event to `video_ended`.

Term	Meanings in code today	Proposed canonical definition Proposed
Survey unlock	Survey window: start + (duration or 90) + 15 min. Access and honorarium: start + (duration or 60), no buffer.	One helper: the Zoom end time if known; otherwise scheduled end + 15 min. Used by every check.
Total earnings	Dashboard Σ PAID versus the `User.totalEarnings` counter.	Σ PAID payments everywhere. Keep the counter as a cache only, or remove it.
Refresh "daily"	YouTube and LinkedIn are on demand. Export ingest runs on weekdays only.	Schedule a daily Content Hub `sync_all_platforms` run (the `next_sync_at` field already assumes a daily sync) and run export ingest every day.

7 Known limitations (consolidated)

ACCURACY (AFFECTS REPORTED NUMBERS)

1. **Report import is all-or-nothing.** One report row for a program discards every webhook or SDK attendee for that program (platform `export.service.ts:496–502`).
2. **Webhook-only attendance has no duration**, so average minutes are understated. LEFT rows are never paired with JOINED rows.
3. **No minimum attendance duration.** A 1-second join verifies attendance and unlocks the honorarium. An in-app SDK join verifies the user's own account.
4. **The Zoom webhook accepts unsigned requests** when the signature header is missing (`zoom-webhook.service.ts:117`). A forged join could verify attendance (inferred).
5. **The packet has no date window.** Registrations include pending and waitlisted.
6. **YouTube views are lifetime**, and "average view duration" is actually video length.
7. **HubSpot email statistics can be multiplied** across campaigns, and the generic key search can double count.
8. **The model writes the narrative numbers** for platform and HubSpot data. They are not computed in code.
9. **Survey completion rate** can understate the true rate or exceed 100%. Survey scores are client-supplied.
10. **Video progress unit mismatch** (0–1 sent, 0–100 documented), so admin progress shows 100× too small.

FRESHNESS

11. Platform connectors are on demand only; Meta, livestream and survey data are uploaded manually.
12. Hub export ingest is not triggered by attendance verification, report imports or survey submissions.
13. Post statistics freshness is owned by ops-console and is not visible to these repos. The catalog adds up to 1 hour of cache.
14. Zoom attendee report import is manual. Its lock is held in memory, so multiple ECS tasks could run the same import twice (inferred).

COVERAGE (METRICS THAT DO NOT EXIST)

15. No engagement, no-show, unique-viewer or watch-time metric.
16. CME credits are never issued (`CMECredit` is empty). Live sessions never mark an enrollment complete.
17. `AnalyticsEvent` and `User.lastLoginAt` are never written, so user activity and recency cannot be measured.
18. Hub's `clip_analytics` and `campaign_analytics` tables have no ingest code. Post `share_count` and `impression_count` are not exposed.
19. Companion telemetry exists only in logs. The knowledge base has no YouTube captions yet.

8 Glossary

Program	A content unit on Platform: an on-demand video program, a WEBINAR, or a MEETING (office hours).
Campaign	A Content Hub campaign. Programs link to it through `Program.campaignId` ; it is the unit of a report.
Registration / Enrollment	Registration is the application and approval record. Enrollment is the access and progress record.
JOINED / LEFT	Zoom participant events stored in `WebinarParticipantEvent` .
Report import	An admin pull of Zoom's official participant report (with durations) into Platform.
Rollup	The export step that combines one person's join rows into a single attendance row.
Verified	Attendance status that unlocks the post-event survey and honorarium.
FEEDBACK / INTAKE	Post-event survey / registration survey.
Input packet	Platform's export of sessions, attendance, registrations and surveys for one campaign.
Export ingest	Content Hub pulling the input packet into its `export_*` tables.
Report packet	Content Hub's combined payload (Platform export, platform data, HubSpot, KOLs, template) consumed by cht-reports.
Platform data	Per-campaign, per-day YouTube, LinkedIn, Meta, livestream and survey rows in Content Hub.
Post	One published copy of a clip on a social platform, holding that platform's counters.
KOL	Key opinion leader. A doctor featured in or relevant to the campaign.
M2M	Cognito client-credentials authentication between Platform and Content Hub.

Native / Computed	Native: a value passed through from a third-party API. Computed: calculated by CHT code.